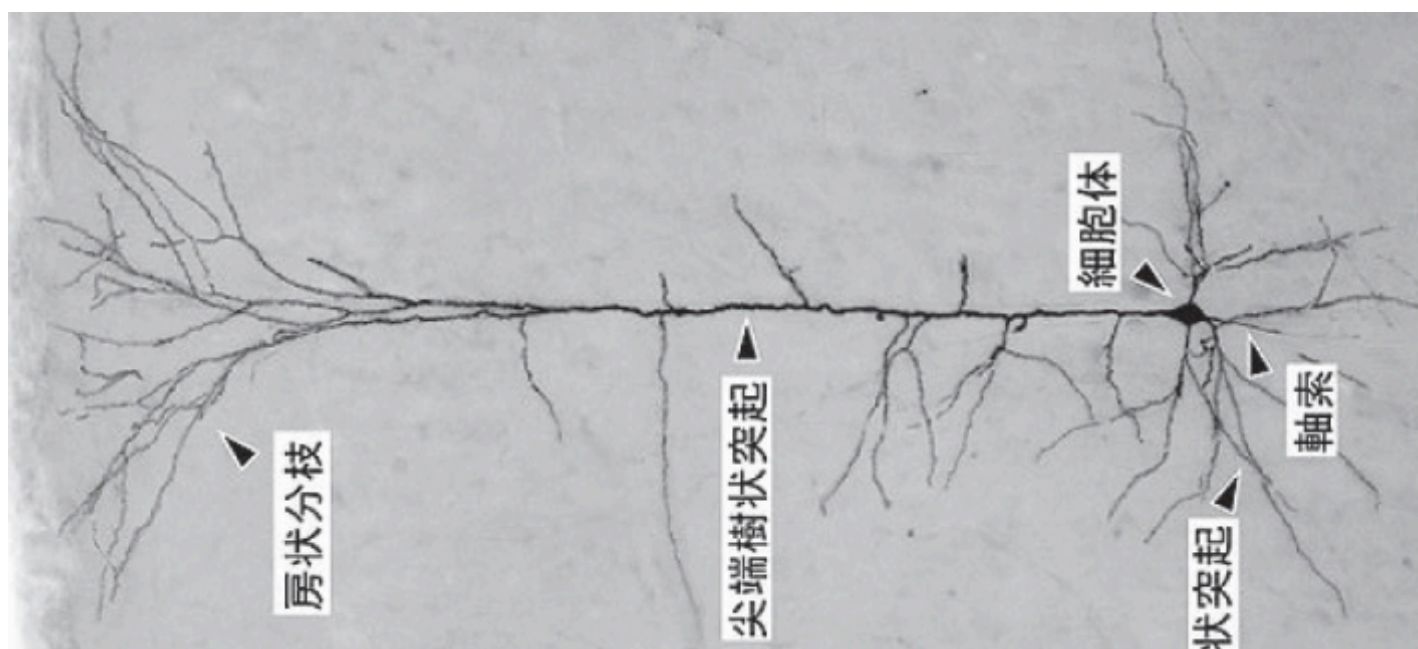


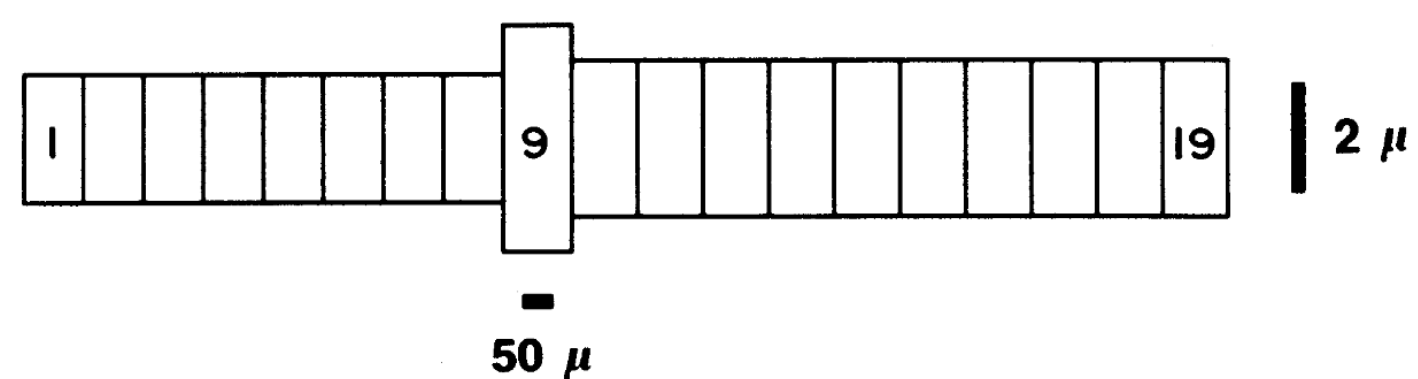
A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

Haruki Munechika

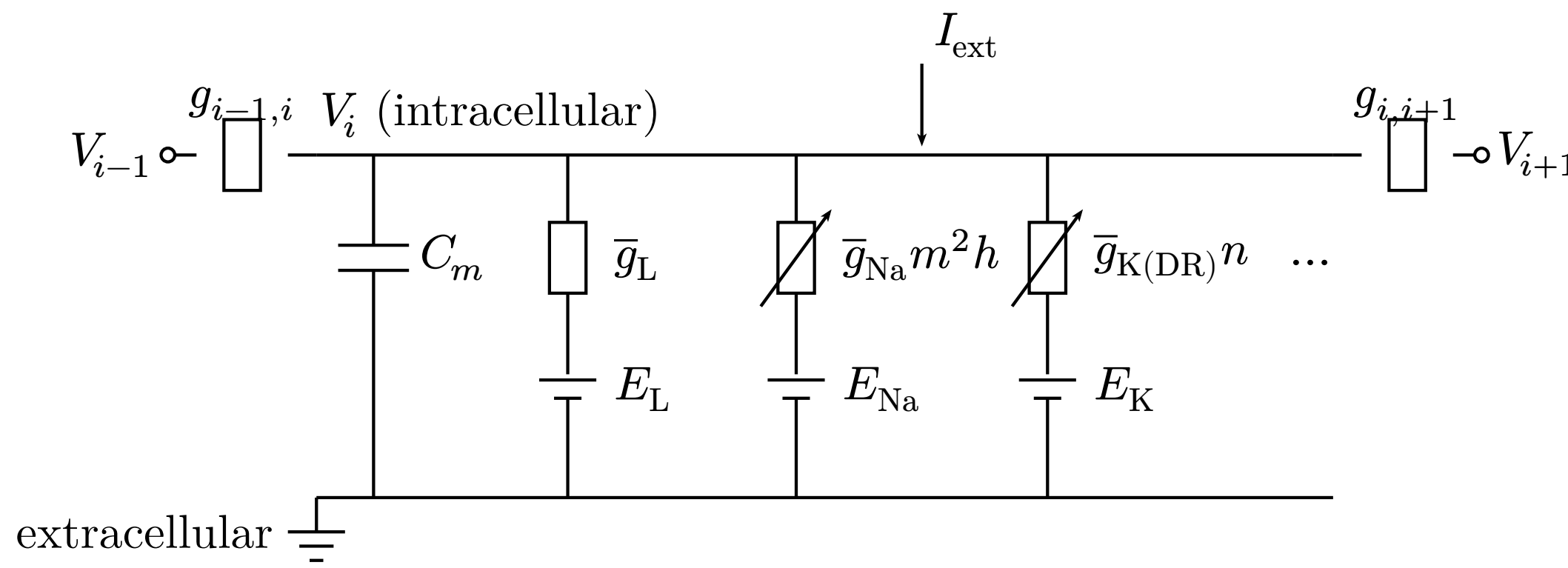
Introduction



CA3 pyramidal neuron [1]



Modelled as 19 compartments [2]



Equivalent circuit of compartment i

Each compartment: capacitance, **variable** ionic conductances, axial coupling $g_{i,i\pm 1}$.

Conductances depend on **gate variables**, ODEs with rate functions α, β :

$$I_{\text{Na}} = \bar{g}_{\text{Na}} m^2 h (V - E_{\text{Na}})$$

$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$

$$\alpha_m(V) = 0.32(13.1 - u)/(\exp((13.1 - u)/4) - 1)$$

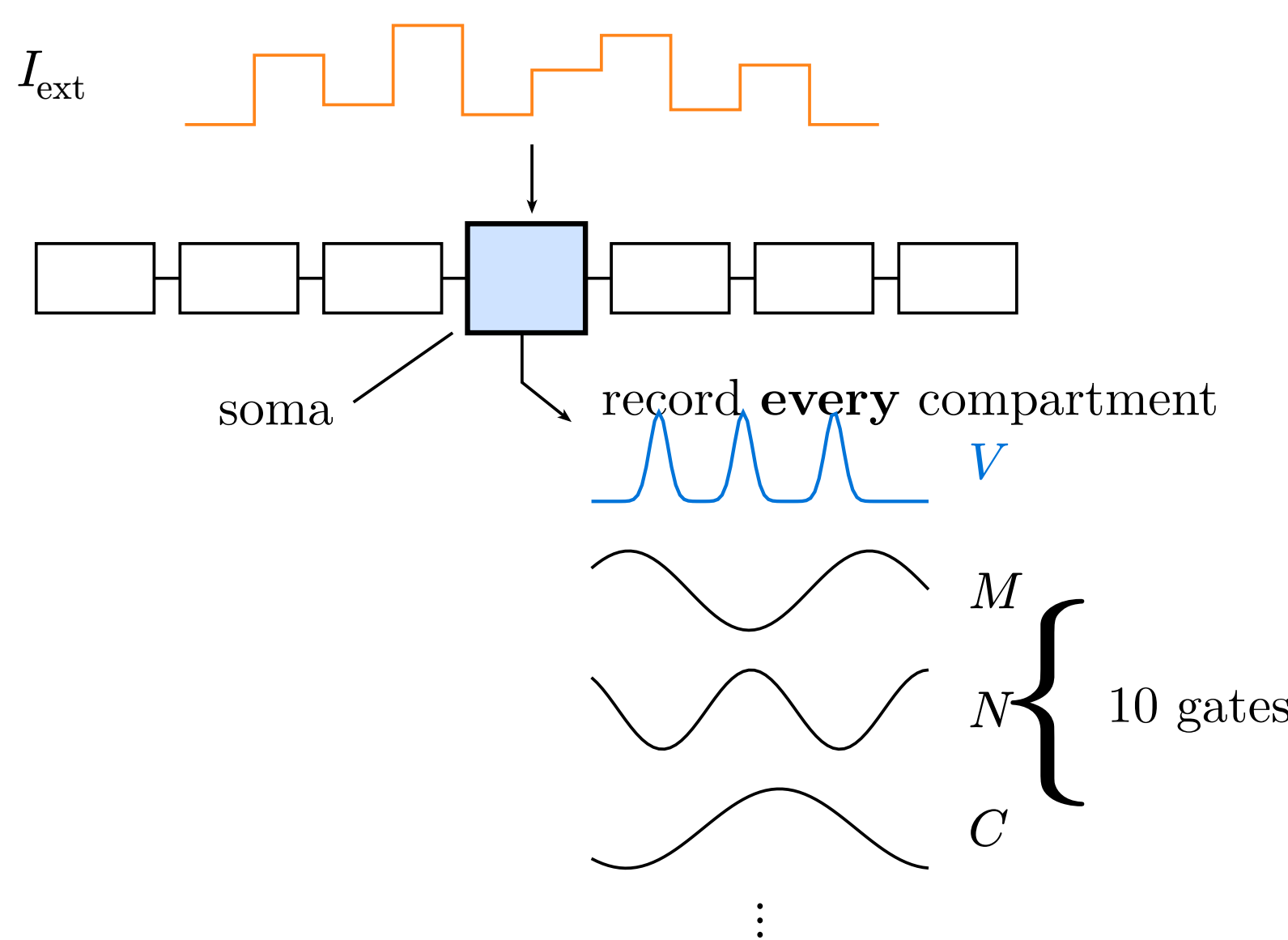
Na current, its gate m , and one rate function ($u = V - E_L$)
→ **11 states per compartment** → **209** for 19; at brain scale the gates become a **memory bottleneck**.

Goal

Reproduce the membrane-potential response with **fewer state variables**.

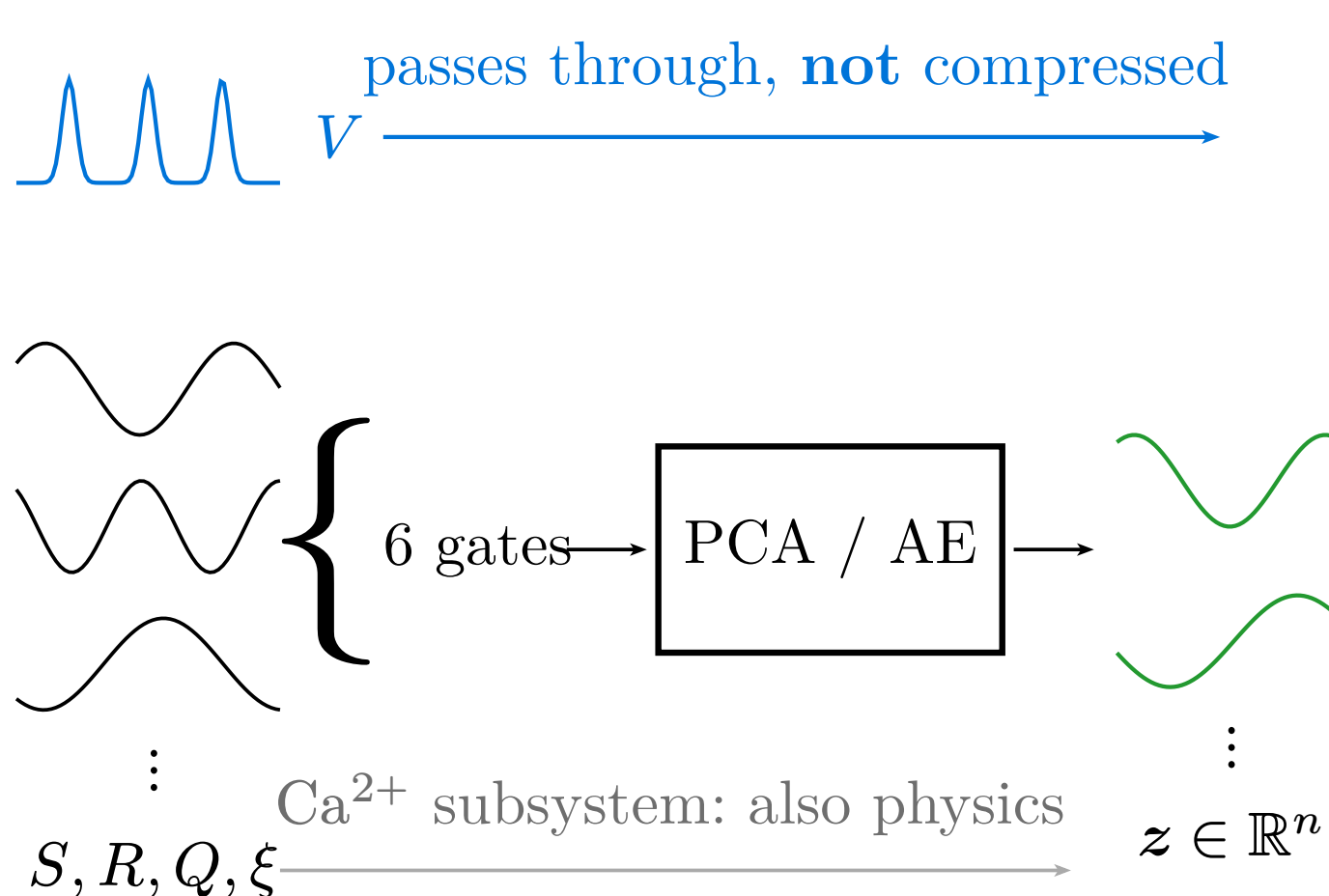
Methods

① Collect the training data



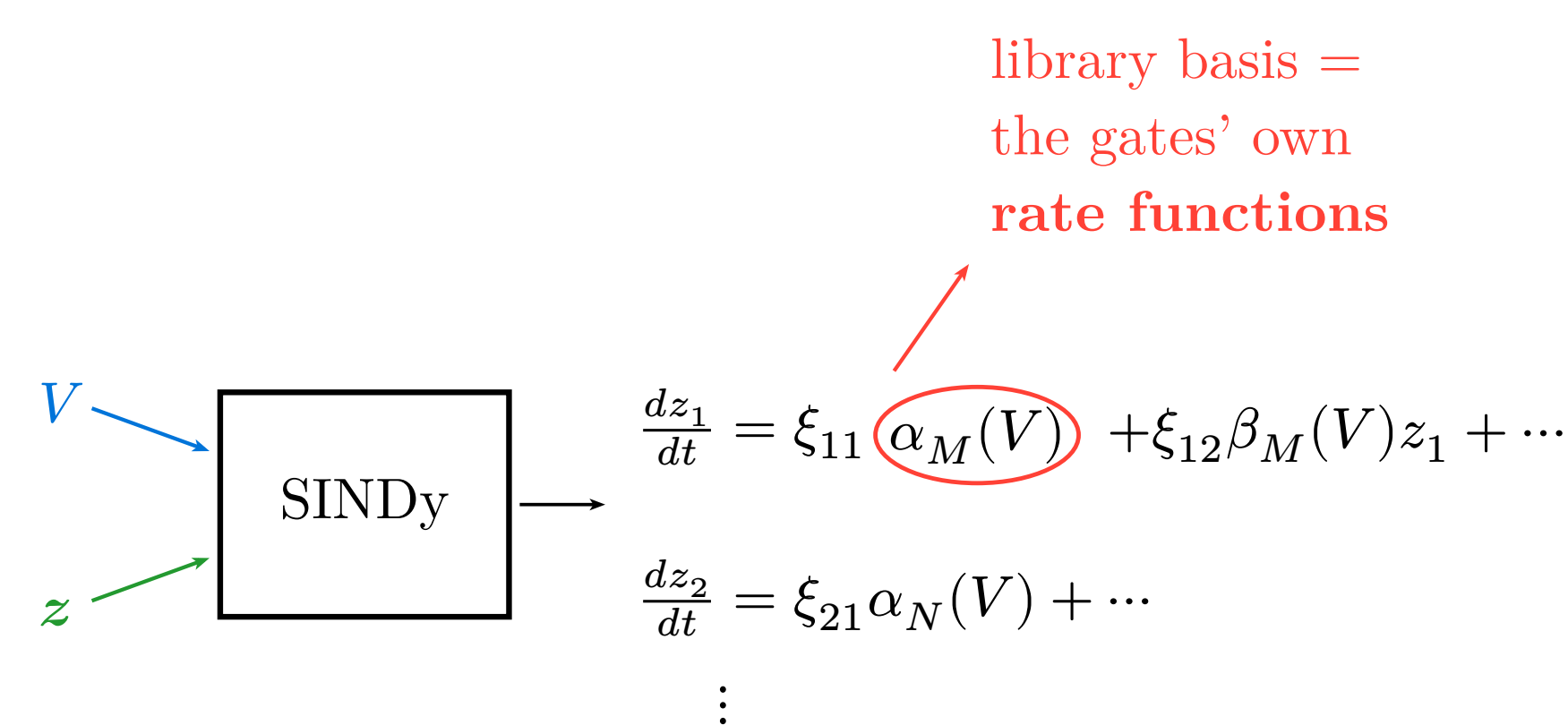
Inject a **random pulse train** at the soma; record V and **10 gates** for all **19 compartments**.

② Compress *only* the gates



6 voltage-dependent gates ride a **low-dimensional manifold** → $z \in \mathbb{R}^n$, $n = 5$. V and Ca^{2+} **stay uncompressed**.

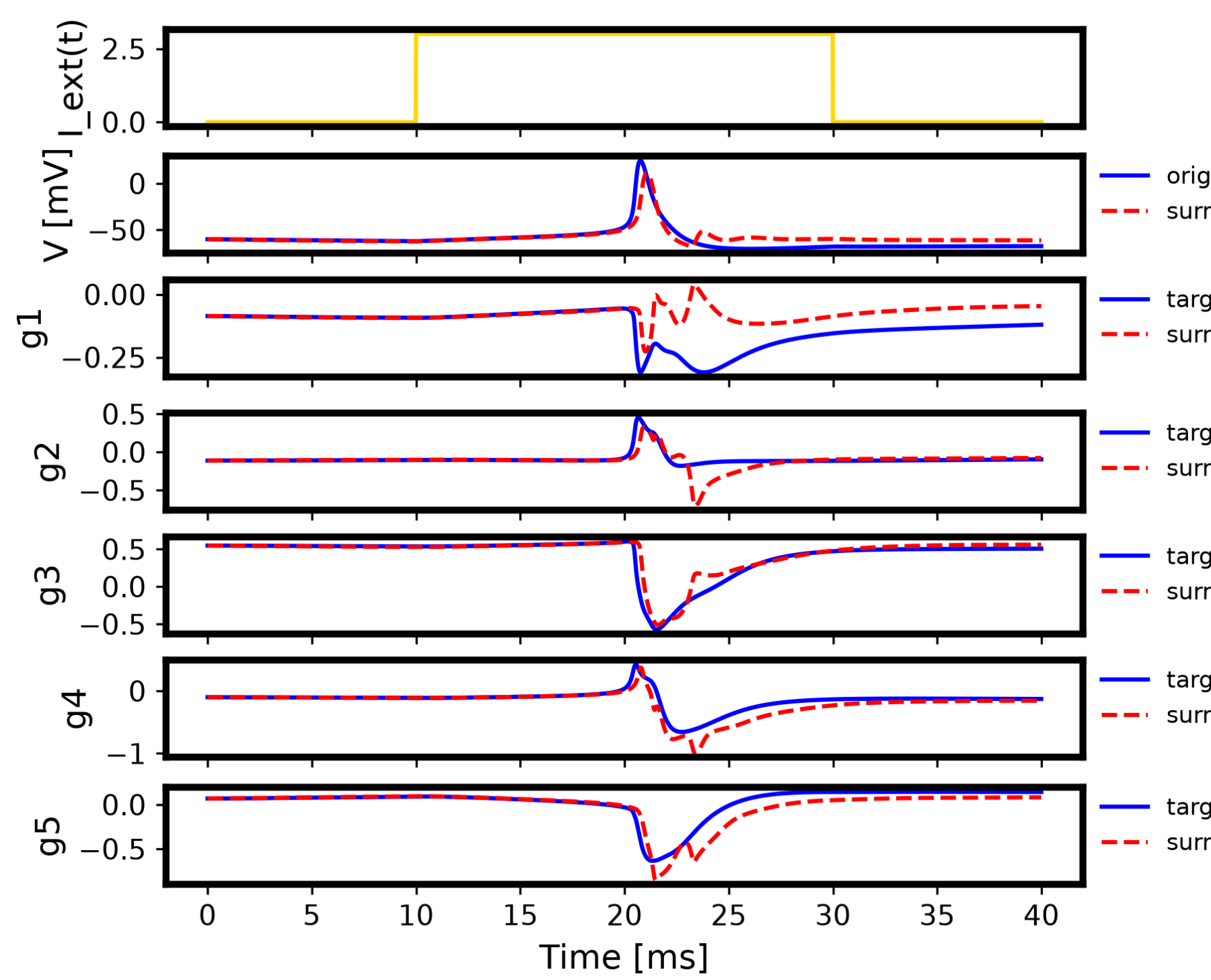
③ Identify the latent ODEs



SINDy [3] fits **only z** ; $\dot{V}$ and Ca^{2+} keep **original physics**. Library is **physics-informed**: gates' own $\alpha(V), \beta(V)$, not a **41-term** generic one.

Results and Discussion

① Single action potential

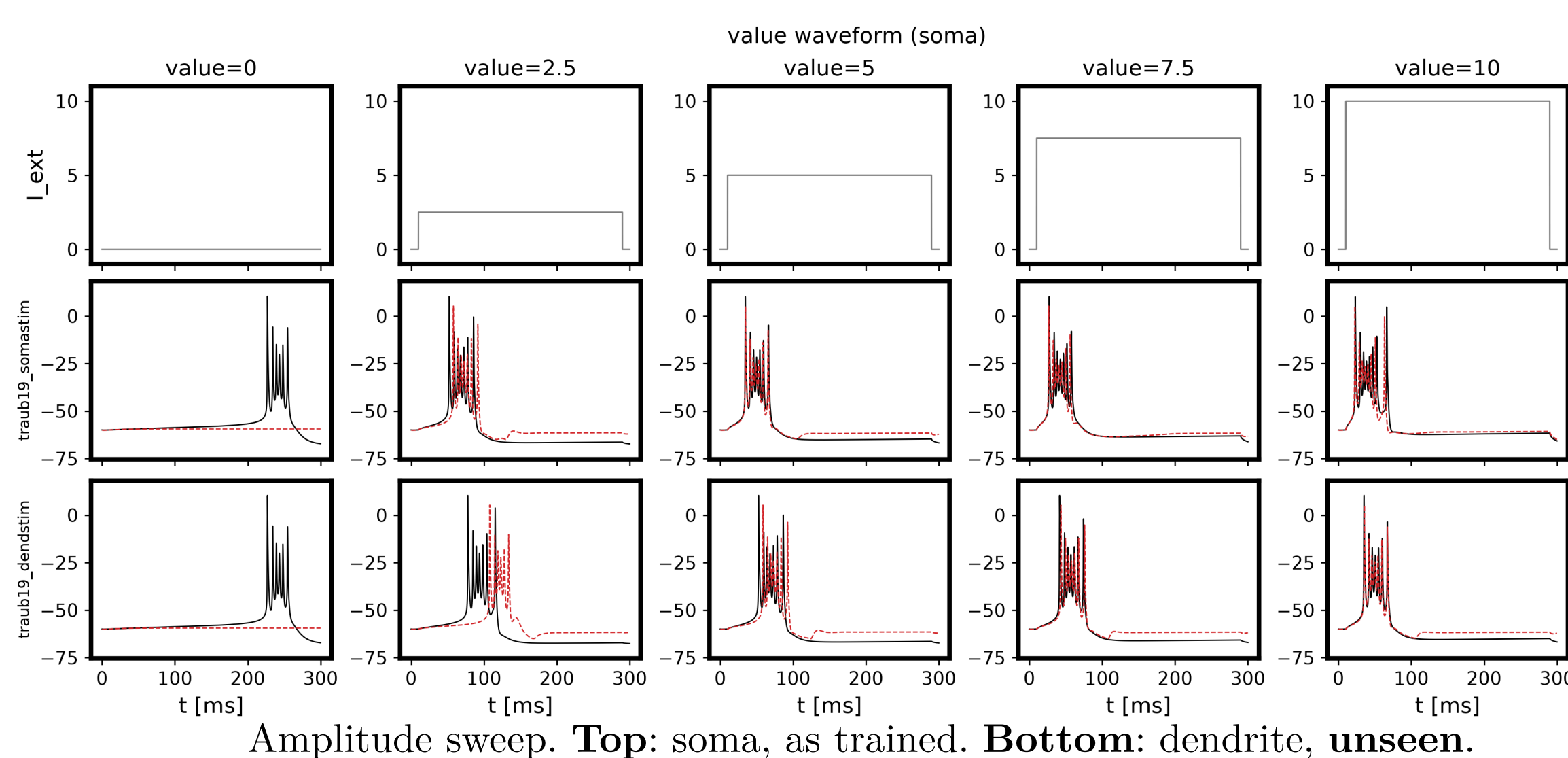


20 ms, 3 $\mu\text{A}/\text{cm}^2$ step: V and the **5 AE latents**. Original (blue) / surrogate (red).

Waveform match

- RMSE **7.3 mV**, shape corr. **0.999**
- spike 1 → 1, latency err **0.3 ms**
- Gap: peak **13 mV** low.

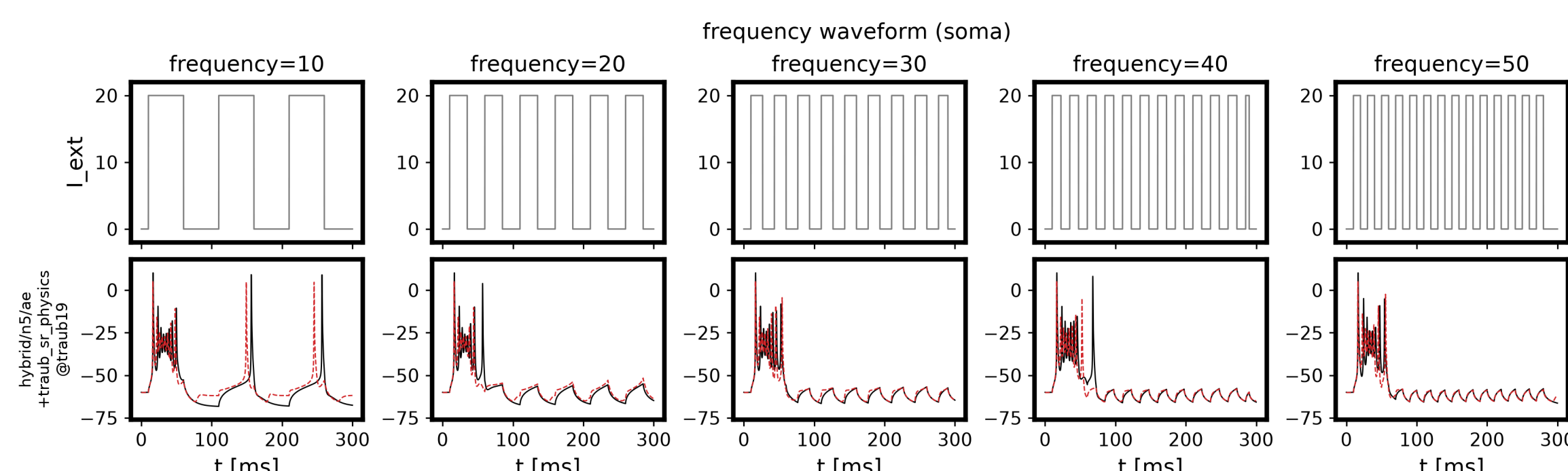
② New stimulus site



Amplitude sweep. **Top**: soma, as trained. **Bottom**: dendrite, **unseen**.

- Bursts reproduced at **both sites** for $I \geq 5$; fires **too early** near threshold.

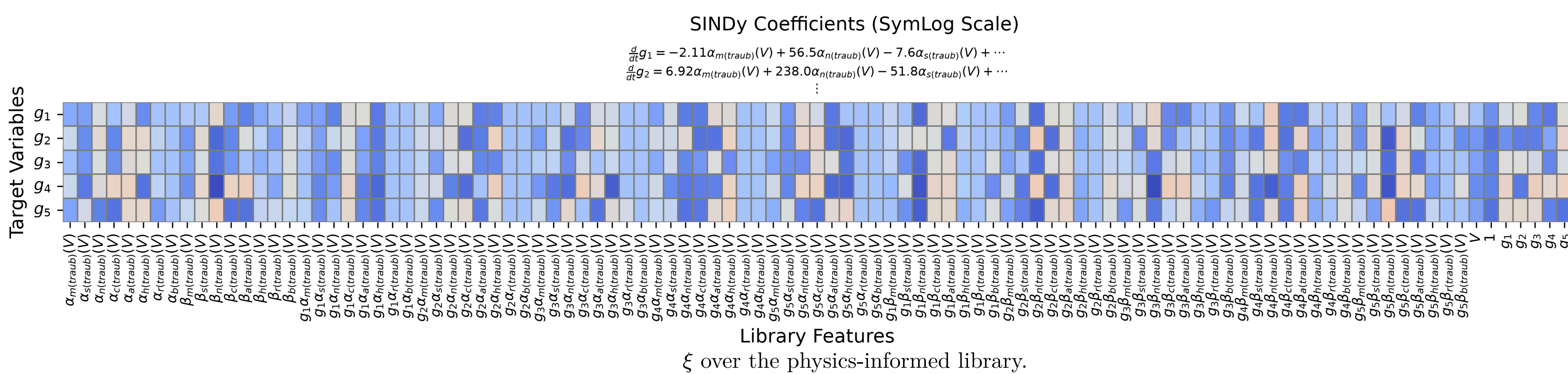
③ Unseen periodic drive



Pulse train, 10–50 Hz — **outside training**.

- Matches for $f \geq 20$ Hz; **drops later spikes** at 10 Hz.

④ Identified latent equations



- 79.6% non-zero** — accurate but **not sparse**: cost **rises** (exp 19 → 121).

Conclusion

- All 19 compartments** replaced, no divergence; AP **shape corr. 0.999**.
- Transfers to an **unseen site and frequency**.
- Compression stops at **6** → **5** and ξ stays dense.
→ **Future**: sparsify ξ , push n down.

Code — <https://github.com/MunechikaHaruki/SINDyNeuroSurrogate>

[1] “錐体細胞.” Accessed: Dec. 08, 2025. [Online]. Available: <https://doi.org/10.14931/bsd.2091>

[2] R. D. Traub, R. K. Wong, R. Miles, and H. Michelson, “A Model of a CA3 Hippocampal Pyramidal Neuron Incorporating Voltage-Clamp Data on Intrinsic Conductances,” *Journal of Neurophysiology*, vol. 66, no. 2, pp. 635–650, Aug. 1991, doi: 10.1152/jn.1991.66.2.635.

[3] K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, “Data-Driven Discovery of Coordinates and Governing Equations,” *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.